INCH-POUND
A-A-59185
27 May 1998

COMMERCIAL ITEM DESCRIPTION

HEATER, HOT WATER BOOSTER, ELECTRIC; SINK, SANITIZING (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers hot water booster heaters that increase a sink's hot water temperature to not less of 180°F. The heaters are intended for use in Naval shipboard food service areas for the sanitizing sink of a three-compartment or four-compartment sink washing operation.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The heater shall have a dual-reservoir system—one to heat water, and the other to trap and collect waste. The waste reservoir shall be able to be drained without draining the heater reservoir. The heater shall be configured for secure mounting to the underside of a sink having a capacity between 51 and 80 gallons. Heaters shall be complete with all electrical wiring and plumbing connections necessary for attaching the heater to the underside of a sink. In the installed condition, the overall height of plumbing connections shall be not greater than 3 inches. Heater dimensions, exclusive of plumbing fittings, shall be not greater than 18 inches long by 9 inches wide by 13 inches high. The heater body and plumbing fittings shall be stainless steel. A quick opening discharge valve shall be provided for draining the water in the sink. The valve shall be rated for operation up to 200°F, and shall be readily accessible to the operator when facing the front of the heater. The heater shall have an ON-OFF switch, an integral temperature indicator to indicate when water reaches the sanitizing temperature, a signal light to indicate when the heater is in operation, and a second signal light to indicate when the high-temperature cutout limit has been exceeded. The heater shall thermostatically control the sink water temperature to settings between 180°F and 195°F. The temperature setting shall be adjustable from the front of the heater. A high temperature cutout shall also be provided to cut power to the heater and prevent damage in the event that there is no water in the heating reservoir, water temperature has reached 205°F, or the regular thermostat fails to operate. The high temperature cutout shall be the manual reset type.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson-Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 7320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

2.2 Operation. Heaters shall raise the water temperature in sinks with volumes of 50 to 78 gallons from 140°F to not less than 180°F, and maintain it at the raised temperature for not less than 4 hours. The heater shall heat 140°F water to not less than 180°F in not greater than 45 minutes.

2.3 Inclined operation. The unit shall operate satisfactorily (in accordance with the requirements of this CID) with no spillage of water or waste when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 6.2)

2.4 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167/1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 6.2)

2.5 Accessibility. The unit shall be physically and visually accessible from the front for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

2.6 Electrical requirements. The unit shall operate on 440 VAC, 60 Hz, single-phase power, as defined in MIL-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not more than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. (Shipboard requirement, see 6.2)

2.7 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 5, UL 174, UL 499, and UL 969.

2.8 Finish. Stainless steel shall have a type 2B or smoother finish, in accordance with ASTM A 480.

2.9 Label plates. The heater shall be provided with a data nameplate, an instruction plate, and a warning plate, each attached to the front of the heater. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the heater.

2.9.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

2.9.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

2.9.3 Warning plate. A warning plate shall be fastened to the heater in a readily visible location with the following information in legible characters:

WARNING
180°F WATER TEMPERATURE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

2.10 High voltage labels. A label reading “Danger—High Voltage” shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger—High Voltage.
- Power supply must be disconnected before servicing.
- Access covers must be in place before use.
- Service should be performed by authorized personnel only.

3. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

5. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.

- Title, number, and date of this CID
- When required, manuals shall be in accordance with ASTM F 760

6.2 Shipboard requirement. Whenever a “(Shipboard requirement)” is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

6.3 Source of documents.

6.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating
MIL-STD-167/1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally excited)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

6.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A 480 - General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM F 760 - Food Service Equipment Manuals

6.3.3 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 5 - Water Heaters, Hot Water Supply Boilers, and Heat Recovery Equipment

6.3.4 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 174 - Household Electric Storage Tank Water Heaters
UL 499 - Electric Heating Appliances
UL 969 - Marking and Labeling Systems

6.3.5 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

6.4 Suggested source of supply. One manufacturer of products known to meet the requirements of this CID is listed below. However, source of supply is not limited to this company, if others are found to meet the requirements of this CID.

Hatco Corporation
635 South 28th Street
Milwaukee, WI 53215-1201

MILITARY INTERESTS:

Military Coordinating Activity

Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity:
Navy - SH
(Project 7320-0958)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.